PHYSICS-INFORMED STATE TRANSITION NETWORK FOR SATELLITE ORBITAL STATE ESTIMATION

Internship Project Presentation

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Learning Orbital Dynamics Beyond Data through Physics-Informed Artificial Intelligence

Existing Approach

🟦 Historical Satellite Data
↓
🟦 Deep Learning Model
↓
🟩 Excellent Short-Term Prediction
↓
🟧 Long-Term Rollout
↓
🟥 Orbit Drift
🟥 Energy Drift
🟥 Physics Violation

Question

Can we design an AI model that predicts satellite state transitions while preserving the fundamental laws of orbital mechanics?

✔️ Energy Conservation
✔️ Angular Momentum Conservation
✔️ Stable Long-Horizon Rollout

Key Challenges

⚡ Physics Preservation

🛰️ Orbital Dynamics

📉 Drift Reduction

🧠 Scientific ML

Accurate prediction is important, but physically consistent prediction is essential for reliable satellite operations

Problem statement

Bridging the Gap Between Data-Driven Learning and Orbital Physics

Existing Limitation

📊Purely Data-Driven Learning-
Neural networks learn statistical patterns but not physical laws.

⚙️ No Physics Constraints
Energy and angular momentum are not explicitly enforced.

⏳ Limited Long-Horizon Stability
Prediction errors accumulate during recursive rollouts.

🌍 Weak Physical Generalization
Models may fit training data but fail to preserve realistic orbital behavior.

Consequences

📉 Orbit Drift
Small prediction errors accumulate over time.

⚡ Energy Drift
Predicted conservation. trajectories

🔄 Angular Momentum Drift
gradually violate energy Orbital plane slowly deviates from the true trajectory.

🚨 Reduced Reliability
Predictions become unsuitable for long-term orbit propagation.

Develop a learning framework that accurately predicts satellite state transitions while preserving the fundamental laws governing orbital motion.

Project Objectives

Overall Objective

Developa Physics-Informed State Transition Network that accurately predicts satellite state transitions while preserving orbital physics.

Specific Objectives

Accurate State Transition Learning
Predict ΔPosition and ΔVelocity with high accuracy.

Long-Horizon Stability
Reduce cumulative orbit drift in recursive rollouts.

Physics-Constrained Learning
Preserve energy and angular momentum during prediction.

Scientific ML Framework
Develop an extensible framework for orbital dynamics learning.

Physics-Based Performance Evaluation
Evaluate using AI metrics and orbital mechanics metrics.

Methodology

Problem Definition
Formulate the satellite state transition learning problem.

Data Preparation
Propagate satellite trajectories using the SGP4 propagator and generate state transition pairs.

Physics Formulation
Compute physics residuals using Newtonian gravitational equations and incorporate orbital dynamics into the learning process.

PI-STN Development
Design a Physics-Informed State Transition Network by integrating the Physics Engine with the neural network.

Model Training
Train using hybrid supervision:Data Loss and Physics Loss.Optimize using Adam + L-BFGS.

Evaluation & Analysis
Evaluate using
RMSE,MAE,R^2,Orbit Rollout,Physics Residuals and Orbital Elements.

The proposed methodology integrates orbital mechanics, synthetic data generation, and Scientific Machine Learning into a unified research framework for satellite state prediction.

System Architecture

🛰️ Orbit Simulation

Purpose: Generate physically accurate training data.

Components:

- TLE Orbital Elements
- SGP4 Orbit Propagation
- Coordinate Transformation (TEME → GCRS → ITRS)
- Synthetic Observation Generation

🧠 Physics-Informed AI

Purpose: Learn satellite state transitions while preserving orbital dynamics.

Components:

- Physics-Informed State Transition Network (PI-STN)
- Physics Engine
- Hybrid Loss Optimization
- Multi-Step Rollout Training

📊 Evaluation

Purpose: Assess prediction accuracy and physical consistency.

Evaluation Metrics:

- RMSE / MAE / R^2
- Position & Velocity Error
- Physics Residual
- Orbit Rollout and Orbital Element Validation.

The proposed architecture integrates orbital physics with deep learning to achieve accurate and physically consistent satellite state prediction.

Hybrid Learning Framework

PI-STN Network

The PI-STN learns satellite state transitions instead of directly predicting future trajectories.

Inputs:

Current State (Position, Velocity) and ΔTime

Learns:

State Transition (ΔPosition, ΔVelocity)

Outputs:

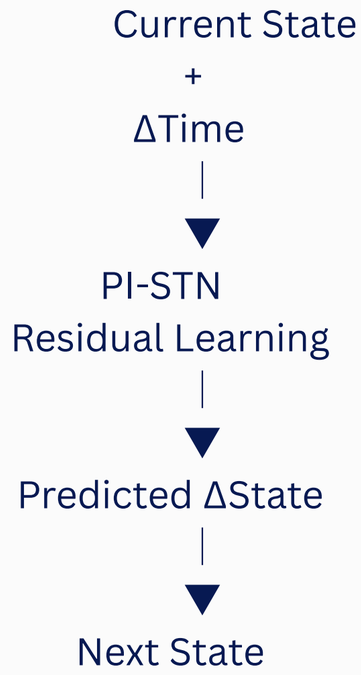
Predicted satellite state increment

Physics Engine

The Physics Engine constrains the neural network using orbital mechanics during training.

- Newtonian gravitational dynamics
- Energy conservation constraint
- Angular momentum conservation
- Physics residual computation
- Hybrid data + physics optimization

Architecture Diagram

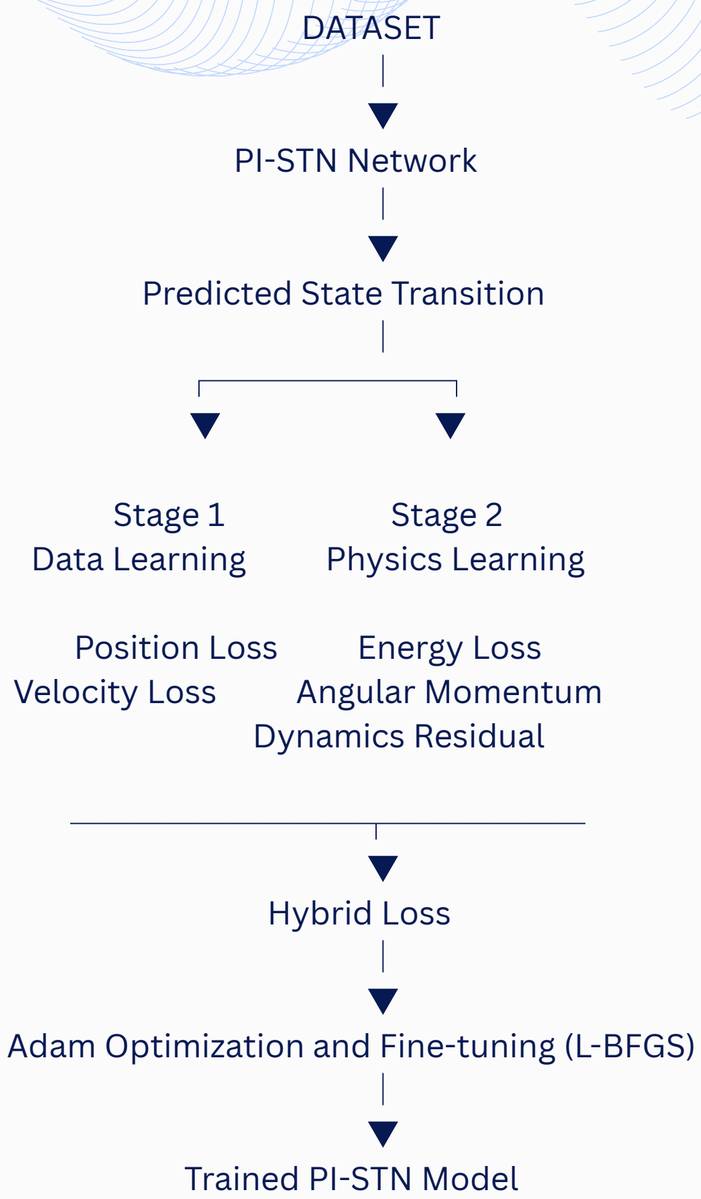


Hybrid Loss = Data Loss + Physics Loss

This enables physically consistent predictions with reduced orbit drift over long propagation horizons.

Dataset & Training Strategy

Training Strategy



Dataset Summary

Satellite : INSAT-3D
Simulation Duration : 7 Days
Sampling Interval : 60 s
Total Samples : 5400

Dataset Split

70% Train
15% Validation
15% Test

Dataset Pipeline

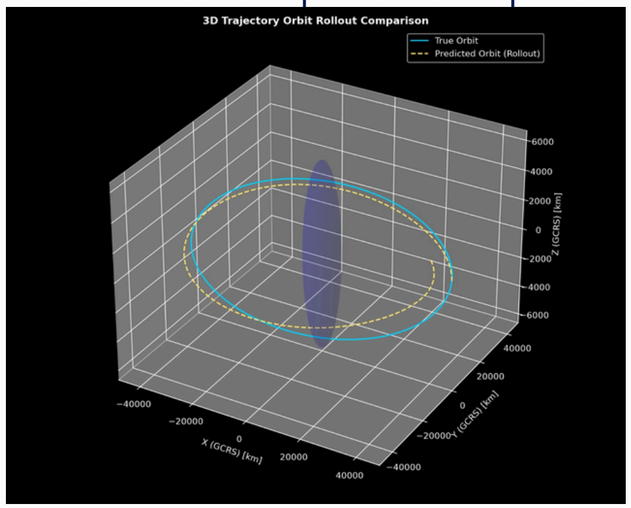
🛰️ Satellite Source
GOES-16 TLE Data
↓
🌍 Orbit Propagation
SGP4 Simulator
↓
📍 Coordinate Transformation
TEME → GCRS → ITRS
↓
📡 Ground Station Simulation
Observation Generation
↓
📊 Final Dataset
State + ΔTime → ΔState

Physics-aware dataset generation combined with hybrid optimization improves long-horizon orbital prediction.

MODEL EVALUATION & RESULTS

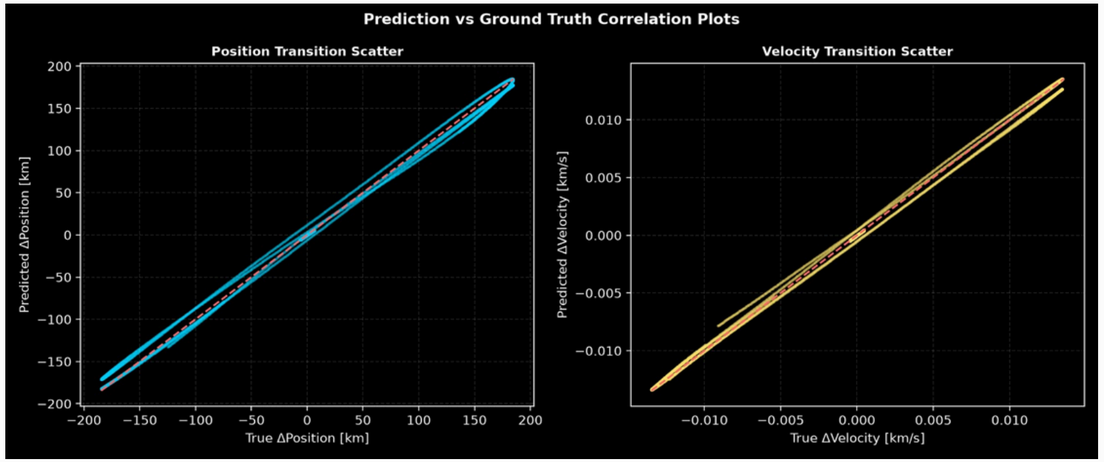
Long-Horizon Rollout

- Predicted trajectory closely follows the true orbit.
- Small cumulative drift appears during recursive rollout.
- Overall orbital shape remains preserved.



Transition Prediction

- Strong agreement in ΔX and ΔY.
- Slight deviation observed in ΔZ.
- Smooth transition learning without high-frequency noise.



Quantitative Metrics

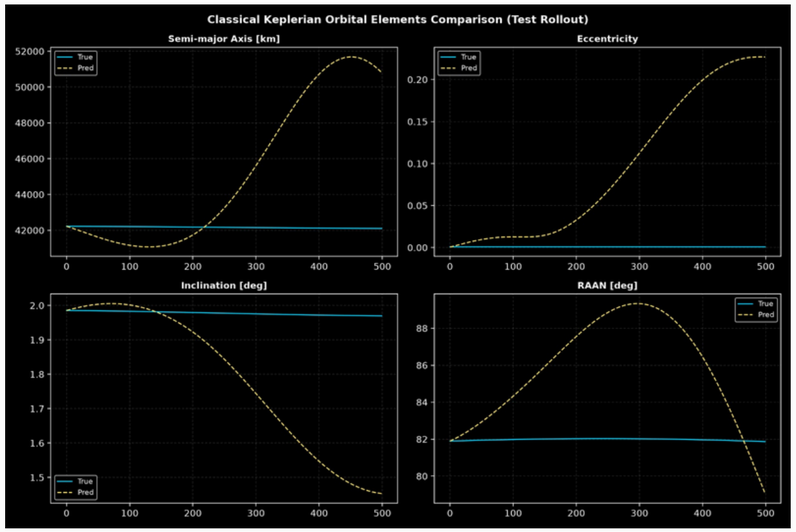
Position RMSE-6.923739 km
Velocity RMSE-0.000387 km/s
Mean Physics Residual-5.6079e-07 km/s²
Mean Rollout Drift-5323.9210 km
R^2 Score-0.995764
MAE-4.948280 km

✓ Low Prediction Error

MODEL EVALUATION & RESULTS

Orbital Validation

- Orbital characteristics remain largely consistent.
- Minor deviations accumulate over long propagation.
- Physics-informed constraints reduce unrealistic trajectory evolution.



Key Findings

- Accurate short-term state transition prediction.
- Stable recursive rollout with limited drift accumulation.
- Physics-informed constraints preserve key orbital characteristics.
- Framework provides a strong foundation for future long-horizon satellite state estimation.

The proposed PI-STN successfully combines data-driven learning with orbital physics, enabling accurate and physically consistent satellite state prediction.

Challenges Faced & Solutions

Long-Horizon Rollout Drift
Challenge:Prediction errors accumulated during recursive state propagation, causing orbit drift.
Solution:Implemented Multi-Step Rollout Training, Curriculum Learning, and Symplectic Integration to improve long-term stability.

Integrating Machine Learning with Orbital Physics
Challenge:Conventional neural networks optimize only prediction accuracy and ignore orbital dynamics.
Solution:Developed a Physics-Informed State Transition Network (PI-STN) by integrating a Physics Engine with hybrid loss functions to enforce orbital constraints during training.

Extensive Research Experimentation
Challenge:Selecting appropriate architectures, rollout strategies, and physics constraints required extensive experimentation.
Solution:Performed iterative ablation studies, hyperparameter tuning, and comparative evaluations to identify the most effective configuration.

Training Stability & Multi-Loss Optimization
Challenge:Balancing transition, physics, residual, energy, and angular momentum losses during optimization.
Solution:Applied hybrid loss weighting, target normalization, Adam → L-BFGS optimization, and early stopping for stable convergence.

The primary challenge was not just predicting satellite motion, but designing a learning framework that combines machine learning with the governing laws of orbital mechanics while maintaining stable long-term predictions.

My Learning Journey

🛰️ Scientific Computing

- Orbital Mechanics
- Numerical Integration (SGP4)
- State Transition Modeling

🧠 AI & Scientific Machine Learning

- Physics-Informed Neural Networks
- State Transition Learning
- Multi-Step Rollout Training
- Hybrid Physics + Data Learning

⚙️ Software Engineering

- Modular Project Architecture
- PyTorch Implementation
- Experiment Management

🔬 Research Skills

- Reading & analyzing research papers & application papers
- Hyperparameter optimization
- Scientific validation and critical analysis
- Experimental design and ablation studies

Theory
↓
Implementation
↓
Experimentation
↓
Scientific Validation
↓
Research Mindset

Research is not about obtaining perfect results,it is about systematically improving ideas..

Future Scope & Research Directions

Future Research Directions

🌐 1. Multi-Satellite Learning

- Extend PI-STN to simultaneously learn from multiple GEO, MEO, and LEO satellites.
- Improve model generalization across different orbital regimes.

🧠 2. Advanced Neural Network Architectures

- Explore Transformer-based PI-STN, Graph Neural Networks.
- Improve long-horizon prediction accuracy and scalability.

⚛️ 3. Adaptive Physics Learning

- Learn unknown perturbations (SRP, atmospheric drag, third-body effects) directly from data.
- Dynamically adjust physics constraints during training.

🚀 4. Digital Twins for Spacecraft

- Develop real-time digital twins for satellite health monitoring and orbit prediction.
- Enable predictive maintenance and autonomous mission planning.

Evolve the proposed PI-STN into a scalable Scientific Machine Learning framework capable of supporting next-generation autonomous satellite operations and Space Situational Awareness.

Research Roadmap

Current PI-STN

▼

Multi-Satellite Learning

▼

Advanced Neural Networks (Transformer / GNN / HNN)

▼

Adaptive Physics Learning

▼

Real-Time Digital Twins

▼

Autonomous Space Mission Support

Conclusion

🎯 Research Objective Achieved

- Developed a Physics-Informed State Transition Network (PI-STN) for satellite state estimation.
- Successfully integrated orbital mechanics with deep learning.
- Established a complete end-to-end Scientific Machine Learning pipeline.

🚀 Major Contributions

- Developed a modular framework from orbit propagation to evaluation.
- Incorporated physics constraints to improve long-horizon prediction stability.
- Enabled prediction of state transitions and residual accelerations.
- Designed a scalable architecture for future multi-satellite learning.

📊 Key Outcomes

- High prediction accuracy ($R^2 > 99.5\%$).
- Improved long-term rollout stability.
- Reduced orbital drift through physics-informed learning.
- Low state prediction error.

Final Outcome

- Combining machine learning with orbital physics enables more reliable, interpretable, and physically consistent satellite orbit prediction than purely data-driven approaches.

The proposed PI-STN demonstrates that Scientific Machine Learning can bridge the gap between classical orbital mechanics and next-generation AI-based satellite state estimation.

References

Astrodynamics & Orbital Mechanics

- Vallado, Fundamentals of Astrodynamics and Applications, 4th Ed.
- Montenbruck & Gill, Satellite Orbits: Models, Methods and Applications
- Curtis, Orbital Mechanics for Engineering Students

Scientific Machine Learning

- Raissi et al. (2019) – Physics-Informed Neural Networks.
- Karniadakis et al. (2021) – Physics-Informed Machine Learning.
- Cuomo et al. (2022) – Scientific Machine Learning through PINNs.

Deep Learning & AI

- Greydanus et al. – Hamiltonian Neural Networks
- Paszke et al. – PyTorch
- Vaswani et al. – Attention Is All You Need

Data Sources & Software

- CelesTrak
- Space-Track
- Astropy
- ISRO (INSAT-3D Mission)
- Scikit-learn

THANK YOU

PHYSICS PROVIDES THE LAWS. AI PROVIDES THE LEARNING. TOGETHER, THEY SHAPE THE FUTURE OF AUTONOMOUS SPACE SYSTEMS.